



Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • Goal: insights to guide strategy

Project & Dataset Overview

Scope

Analyze spending, segments, product preferences, subscriptions

Dataset

Rows: 3,900 • Columns: 18 •
Missing: 37 review_rating values

Key Features

Demographics, purchase details, behavior, subscription status



Methodology — Python & SQL



01

Python EDA

Data exploration, null checks, imputation

02

Feature Engineering

Age groups, frequency metrics, formatting

03

SQL Analysis

Revenue & product rankings, segmentation



Revenue Highlights

75,191

Female Revenue

157,890

Male Revenue

Male customers generated higher total revenue in the dataset

Product & Rating Insights



Top products by average rating

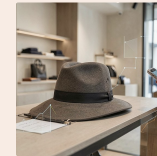
- Gloves – 3.86
- Sandals – 3.84
- Boots – 3.82
- Hat – 3.80
- Skirt – 3.78

📋 Use top-rated items for promotional placement

Discounts & High-Spend Users



839 customers used discounts yet spent above average purchase amount



High discount rate items

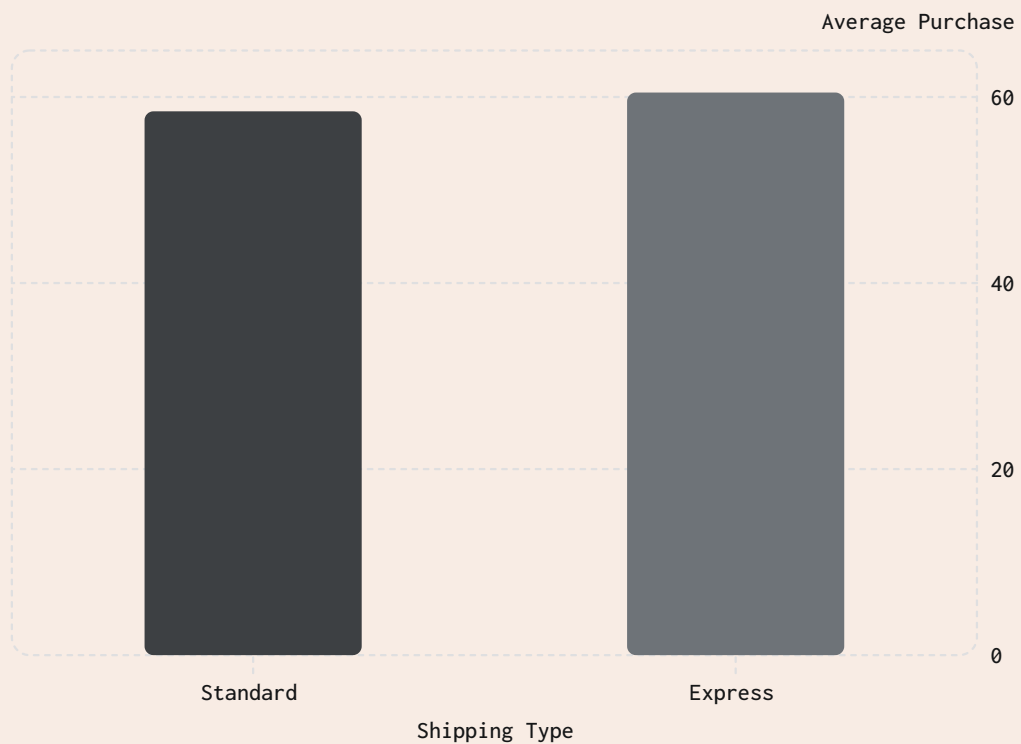
Hat – 50% discounted purchases



Sneakers

49% discounted purchases

Shipping & Spending



Express shipping customers spend slightly more on average



Subscription & Repeat Buyers



1,053

Subscribers

Average spend $\approx$ \$59.49 • Total revenue
62,645



2,847

Non-subscribers

Average spend $\approx$ \$59.87 • Total revenue
170,436



958

Repeat buyers (>5)

958 subscribed, 2,518 non-subscribed

Customer Segments & Top Categories



Loyal

3,116 customers



New

83 customers



Returning

701 customers

Top items by category (sample)

- Accessories: Jewelry
- Clothing: Blouse
- Footwear: Sandals



Recommendations & Next Steps

Boost Subscriptions

Promote exclusive subscriber benefits

Reward Loyalty

Build loyalty programs for repeat buyers

Review Discount Policy

Monitor high-discount items for margin control

Product Positioning

Feature top-rated items in campaigns

Targeted Marketing

Focus on high-revenue customer segments

